



Royal University of Phnom Penh

Faculty of Engineering

Department of Information Technology Engineering

SUBJECT : APL

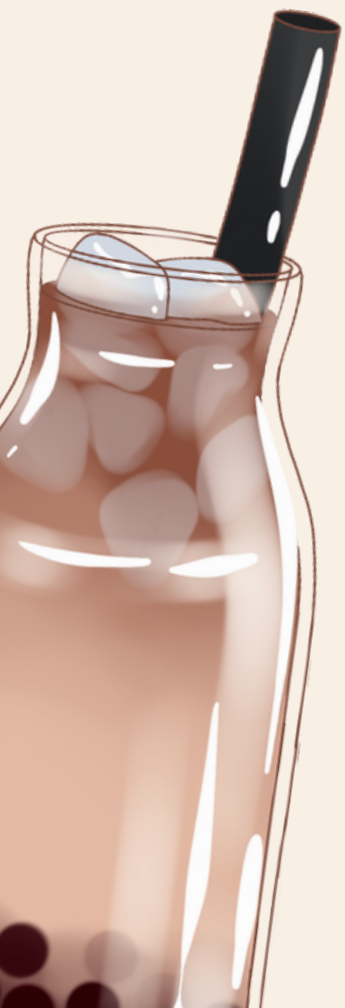
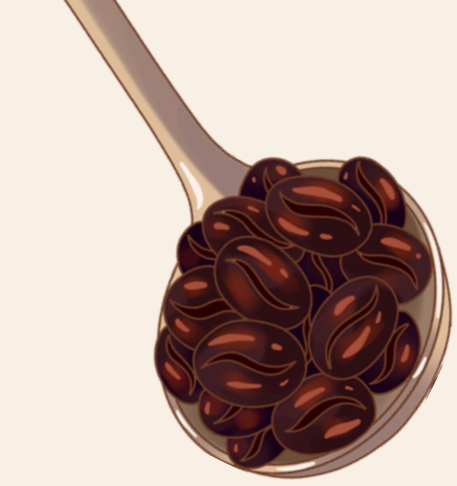
COFFEE ORDERING SYSTEM

LECTURE : SAKTIKA



GROUP MEMBERS

Po Phearun
Nim Cheyseth
Thet Englang
Tep Panhasak
Phal Phanna



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1. Problem Statement

- Inefficient Order Tracking & Human Error
- Chaotic Queue Management
- Lack of Security and Access Control
- Financial Blind Spots for Ownership



2.Solution

_Centralized Digital Ordering

- Replace verbal and handwritten tickets with a strict digital cart system..

_Dynamic Order State Machine

- Digital queue tracking system for the barista station.

_Strict Role-Based Access Control (RBAC)

- Implement unified authentication to lock down sensitive backend routes and database tables.

_Automated Financial Dashboards

- Separate the payment processing logic from the order creation to capture every successful transaction securely.



3. System Functions

Admin Role (Shop Owner)

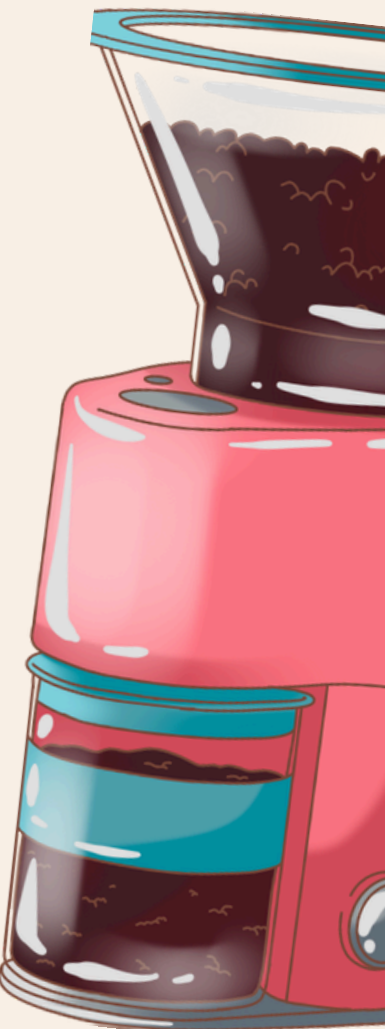
- Add, update, and delete menu categories (e.g., Hot Drinks, Frappes, Pastries).
- Create new menu items and strictly assign them to an existing category upon creation.
- View daily sales dashboards, top-selling items, and peak business hour reports.
- Update base prices, item details, and toggle availability for out-of-stock items.
- Delete discontinued drinks entirely from the system.

Customer Role (Optional)

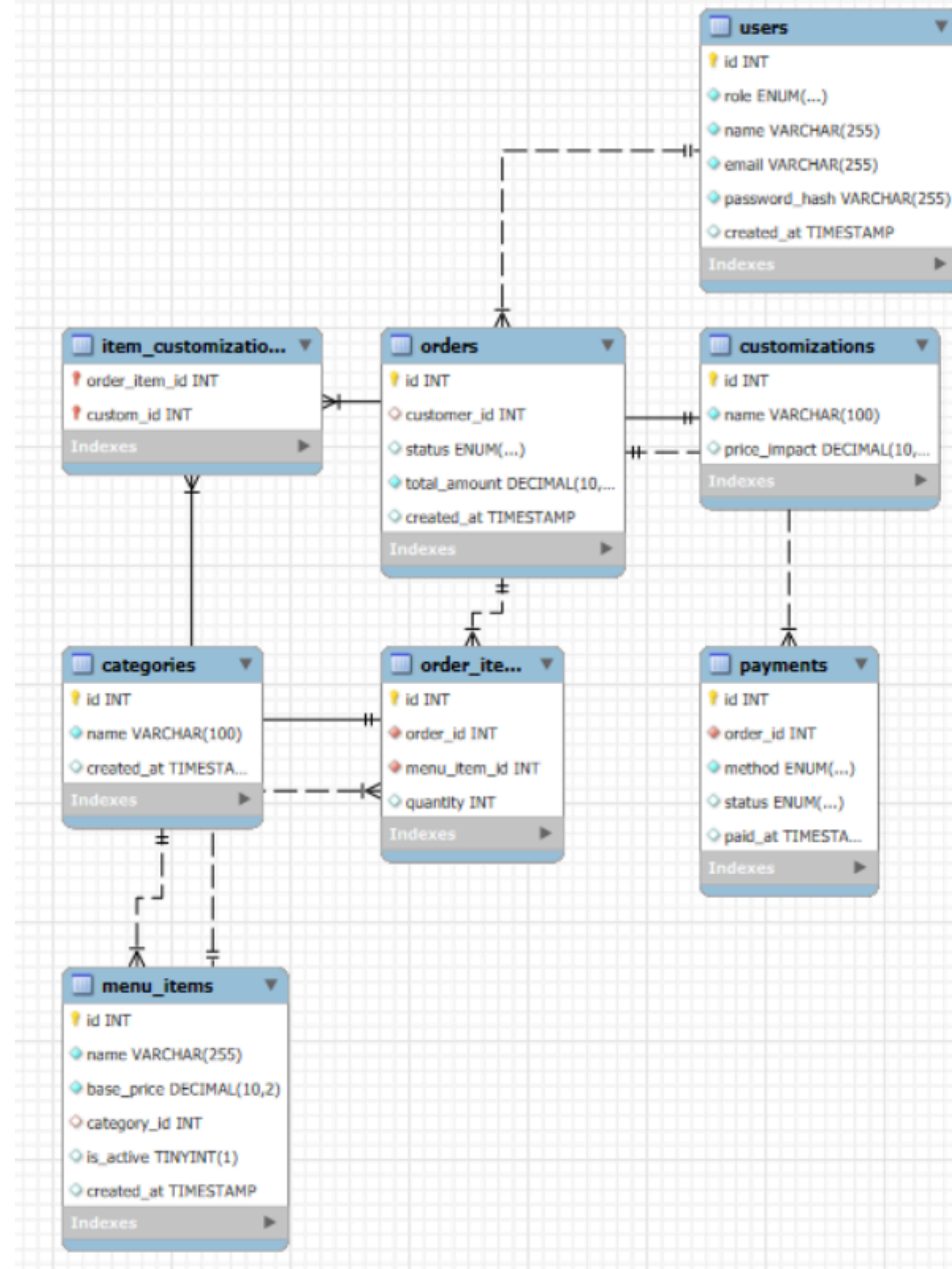
- View the digital menu catalog to browse available items (Strictly read-only access).
- Place remote orders by adding items to a digital cart and checking out.
- Track their live order status remotely to see if their drink is ready for pickup yet.

Staff Role (Cashier / Barista)

- Create new orders by selecting items and applying specific customer customizations.
- Process financial transactions (Cash, Card, QR Code) and generate receipts.
- View the real-time order queue to see exactly which pending drinks need to be brewed next.
- Update order statuses manually to push them from Pending to Brewing to Completed.



4. EER Diagram



THANK
YOU

